A SMOOTH PROPER CURVE OF GENUS AT LEAST 2 HAS A CLOSED POINT OF SEPARABLE DEGREE AT MOST 2G - 2

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the perfect-case argument, the reduction to infinite characteristic p, the Frobenius-twist exclusion of possibility (2), and the closing separability and degree bounds over the etale locus are all retained verbatim.

Editorial note. The label lemma-point-over-separable-extension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The canonical linear series and maps to the projective line. Same chapter, Lemmas lemma-genus-smooth, lemma-linear-series-trivial-existence and lemma-g1d. Gives deg(omega) = 2g - 2 with g global sections, extracts a g^1_{2g-2} from the g^{g-1}_{2g-2}, and converts it into a morphism X -> P^1_k of degree d <= 2g - 2.
- Finiteness and degree bookkeeping. Same chapter, Lemmas lemma-flat and lemma-finite; Morphisms, Lemmas morphisms-lemma-finite-flat, morphisms-lemma-finite-locally-free-universally-bounded, morphisms-lemma-characterize-universally-bounded and morphisms-lemma-etale-over-field. Makes the morphism finite locally free of degree d, bounds residue-field degrees by d, and gives separability of kappa(u)/kappa(t) over the etale locus.
- Char p control via the Frobenius twist. Same chapter, Lemmas lemma-grd-inequalities and lemma-inseparable-linear-system. Shows X^{(p)} carries no g^{g-1}_d for d <= g - 1, and excludes possibility (2) of the inseparable-linear-system dichotomy, producing a generically etale map of degree <= 2g - 2.

Source reference. Original reference 08YX: sources/stacks-curves/pinned-original-source.tex, source lines 323–323. Prerequisite pointer invoked at proof line(s) 2990; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00HD: sources/stacks-curves/pinned-original-source.tex, source lines 303–303. Prerequisite pointer invoked at proof line(s) 2989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0CCR: sources/stacks-curves/pinned-original-source.tex, source lines 626–626. Prerequisite pointer invoked at proof line(s) 2984; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1A: sources/stacks-curves/pinned-original-source.tex, source lines 1760–1760. Prerequisite pointer invoked at proof line(s) 2980; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CCS: sources/stacks-curves/pinned-original-source.tex, source lines 660–660. Prerequisite pointer invoked at proof line(s) 3012; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CD3: sources/stacks-curves/pinned-original-source.tex, source lines 2915–2915. Prerequisite pointer invoked at proof line(s) 3013; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CCQ: sources/stacks-curves/pinned-original-source.tex, source lines 613–613. Prerequisite pointer invoked at proof line(s) 2982; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03J5: sources/stacks-chow/tags.txt, source lines 4627–4627. Prerequisite pointer invoked at proof line(s) 3001; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Prerequisite pointer invoked at proof line(s) 3026; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Prerequisite pointer invoked at proof line(s) 2992; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CC2: sources/stacks-chow/tags.txt, source lines 15492–15492. Prerequisite pointer invoked at proof line(s) 3000; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a smooth proper curve over k with $H^0(X, \mathcal{O}_X) = k$ and genus $g \geq 2$. Then there exists a closed point $x \in X$ with $\kappa(x)/k$ separable of degree $\leq 2g - 2$.*

Proof. Set $\omega = \Omega_{X/k}$. By Lemma 0C1A this has degree $2g - 2$ and has g global sections. Thus we have a $\mathfrak{g}^{g-1}_{2g-2}$. By the trivial Lemma 0CCQ there exists a $\mathfrak{g}^1_{2g-2}$ and by Lemma 0CCR we obtain a morphism

$$\varphi : X \longrightarrow \mathbf{P}^1_k$$

of some degree $d \leq 2g-2$. Since φ is flat (Lemma 00HD) and finite (Lemma 08YX) it is finite locally free of degree d (Morphisms, Lemma 02KB). Pick any rational point $t \in \mathbf{P}^1_k$ and any point $x \in X$ with $\varphi(x) = t$. Then

$$d \geq [\kappa(x) : \kappa(t)] = [\kappa(x) : k]$$

for example by Morphisms, Lemmas 0CC2 and 03J5. Thus if k is perfect (for example has characteristic zero or is finite) then the lemma is proved. Thus we reduce to the case discussed in the next paragraph.

Assume that k is an infinite field of characteristic $p > 0$. As above we will use that X has a $\mathfrak{g}^{g-1}_{2g-2}$. The smooth proper curve $X^{(p)}$ has the same genus as X. Hence its genus is > 0. We conclude that $X^{(p)}$ does not have a $\mathfrak{g}^{g-1}_d$ for any $d \leq g-1$ by Lemma 0CCS. Applying Lemma 0CD3 to our $\mathfrak{g}^{g-1}_{2g-2}$ (and noting that $2g-2/p \leq g-1$) we conclude that possibility (2) does not occur. Hence we obtain a morphism

$$\varphi : X \longrightarrow \mathbf{P}^1_k$$

which is generically étale (in the sense of the lemma) and has degree $\leq 2g-2$. Let $U \subset X$ be the nonempty open subscheme where φ is étale. Then $\varphi(U) \subset \mathbf{P}^1_k$ is a nonempty Zariski open and we can pick a k-rational point $t \in \varphi(U)$ as k is infinite. Let $u \in U$ be a point with $\varphi(u) = t$. Then $\kappa(u)/\kappa(t)$ is separable (Morphisms, Lemma 02GL), $\kappa(t) = k$, and $[\kappa(u) : k] \leq 2g-2$ as before. □